Shravan Paladugula

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EDUCATION

University of California, San Diego — B.S. Computer Engineering, Jacobs School of Engineering — 2026 – 2030 (expected)

Mountain House High School — Graduated 2026

GPA 4.64 · ranked 6 of 615 · Distinguished Award (top 2%) · AP Scholar with Distinction

EXPERIENCE

Founder & Engineer — Central Valley Printing Company — 2022 – 2025

- Ran a production 3D printing studio to $3,000+ in revenue, owning the full CAD-to-delivery pipeline.
- Engineered 20+ custom parts for 50+ students competing nationally in FTC, DECA, TSA, and SkillsUSA.
- Built and shipped the storefront in TypeScript, React, Vite, Tailwind, and Supabase — live quoting and automated order tracking across 22 production deployments.

SELECTED PROJECTS

Rosetta MD — handheld clinical-speech translator — 2026

Best Hardware Hack, CruzHacks 2026 (278 competitors) · team of 4 · ESP32, MEMS mic, LiPo, OLED, RFID, FastAPI, React

- Owned the hardware: 3.7V LiPo with charge and over-discharge protection, printed enclosure, OLED device UI, and RFID patient check-in.
- Kept power, audio, display, and the AI pipeline coordinated inside one handheld shell, built in 36 hours.

CONJR — voice-to-fabrication pipeline — 2026

You.com Best Project, WizardHacks 2026 · Python, FastAPI, OpenSCAD, CuraEngine, Moonraker, Orange Pi 5 Pro, Three.js

- Built an end-to-end pipeline turning a spoken part request into parametric OpenSCAD, then STL, slice, and gcode on a live printer.
- Added a structural-integrity agent that grades every design OK / WARN / FAIL and blocks unsafe geometry from reaching the machine.

PillWatch — authenticated medication dispenser — 2025

Best Hardware Hack, CruzHacks 2025 (225 competitors) · team of 4 · Raspberry Pi, ESP32, servos, 3D printing

- Designed the dispensing mechanism — custom pinwheel geometry and dual-servo timing tuned to release exactly one dose per rotation.

CHEHRA — face-recognition classroom check-in — 2024 – present

Team of 4 · Python, OpenCV, FastAPI, Raspberry Pi, Arduino, servos, ultrasonic

- Owned hardware integration: servo camera mount that adjusts to student height, ultrasonic approach detection, and the serial layer driving LCD, LED, and buzzer feedback.

UAS-SAR — synthetic-aperture radar drone — 2025

MIT Beaver Works Summer Institute · 160+ engineering hours · MATLAB, Python, C++

- Built a custom airframe and electronics with a SAR payload; implemented PID flight control, SAR backprojection, and a real-time visualization GUI.

LEADERSHIP

President & Lead Director — MountainHacks / MHHS Hackathon Club — 2023 – 2026

Grew to 500+ members; directed the Central Valley's largest student hackathon (200+ attendees, $150K prizes, GitHub and AoPS sponsors); raised $20K for STEM programming.

Co-Founder & Trustee — ServingHandz, national nonprofit — 2021 – 2026

Scaled to 45,000+ meals, 35 chapters, and 1,600+ volunteers; built the legal and financial infrastructure from scratch.

Lead Programmer — VEX Robotics 19359 & FTC 23649 — 2022 – 2026

Led autonomous routines, sensor fusion, and pneumatics; qualified for California state competition in both programs.

SKILLS

Embedded	ESP32, Arduino, Raspberry Pi, Orange Pi, KiCAD, Fusion 360, 3D printing, UART / I2C / SPI, PWM / PID, LiPo power
Languages	C / C++, Python, TypeScript, MATLAB, Embedded Linux, OpenSCAD
Software	React, Next.js, FastAPI, Node.js, Supabase, OpenCV, Three.js, Git / GitHub Actions, Vercel

AWARDS

9 winning projects across 15 hackathons · CruzHacks Best Hardware Hack 2026 & 2025 (back to back) · WizardHacks 2026 You.com Best Project · CruzHacks 2023 Finalist · DECA California State Champion, ICDC top 20 · TSA Engineering Design State Champion & National Qualifier · Oracle Java Certified · President's Volunteer Service Award, Gold (300+ hours)